Lab 1

MSDS684_S70_Reinforcement Learning

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March 2026

Project Overview

This lab investigates the exploration-exploitation tradeoff in the multi-armed bandit problem, one of the simplest reinforcement learning settings described in Sutton and Barto (Chapter 2). In this problem, an agent repeatedly selects from a fixed set of actions (arms), each associated with an unknown reward distribution, with the goal of maximizing cumulative reward over time. The challenge lies in balancing exploration (trying new actions to gain information) and exploitation (choosing the best-known action).

In this implementation, a 10-armed bandit environment was created using Gaussian reward distributions, where each arm has a true mean sampled from a normal distribution. The agent does not have access to these true values and must estimate them through interaction. The environment does not include state transitions, making it a stateless problem where each action yields an immediate reward independent of previous actions.

Two action-selection strategies were implemented and compared: ε-greedy and Upper Confidence Bound (UCB). The ε-greedy method selects a random action with probability ε to encourage exploration, and otherwise selects the action with the highest estimated value. In contrast, UCB incorporates uncertainty directly into action selection by adding a confidence bonus to less-explored actions, encouraging more directed exploration.

Although the bandit problem does not fully represent a Markov Decision Process (MDP), it can be viewed as a simplified case where there is a single state. In this context, the action space consists of the 10 available arms, the reward function is defined by the Gaussian distributions, and transitions are trivial since the state does not change. This connects to the broader MDP framework introduced in Sutton and Barto (Chapter 3), where decision-making extends across states and time.

The expected behavior is that UCB will outperform ε-greedy in terms of both average reward and optimal action selection, due to its more efficient exploration strategy. Additionally, different ε values are expected to demonstrate the tradeoff between exploration and exploitation, where higher ε leads to more exploration but potentially slower convergence to the optimal action.

Deliverables

GitHub Repository:

https://github.com/MPalacios-Analytics/DataScience_Projects/tree/a85d7994c79e39f831d5a655ad960026ea4006c4/06_Reinforcement_Learning/Lab_1

Implementation Summary

This project implements a custom 10-armed bandit environment along with two action-selection algorithms: ε-greedy and Upper Confidence Bound (UCB). The environment generates rewards from Gaussian distributions with unknown means, requiring the agent to learn optimal actions through interaction.

Experiments were conducted over 2000 time steps and averaged across 1000 independent runs to reduce variance. For ε-greedy, three exploration rates were tested (ε = 0.01, 0.1, 0.2), while UCB was evaluated with different confidence parameters (c = 1, 2, 5). The performance of each method was measured using two key metrics: average reward over time and the percentage of optimal action selections.

Key Results & Visualizations

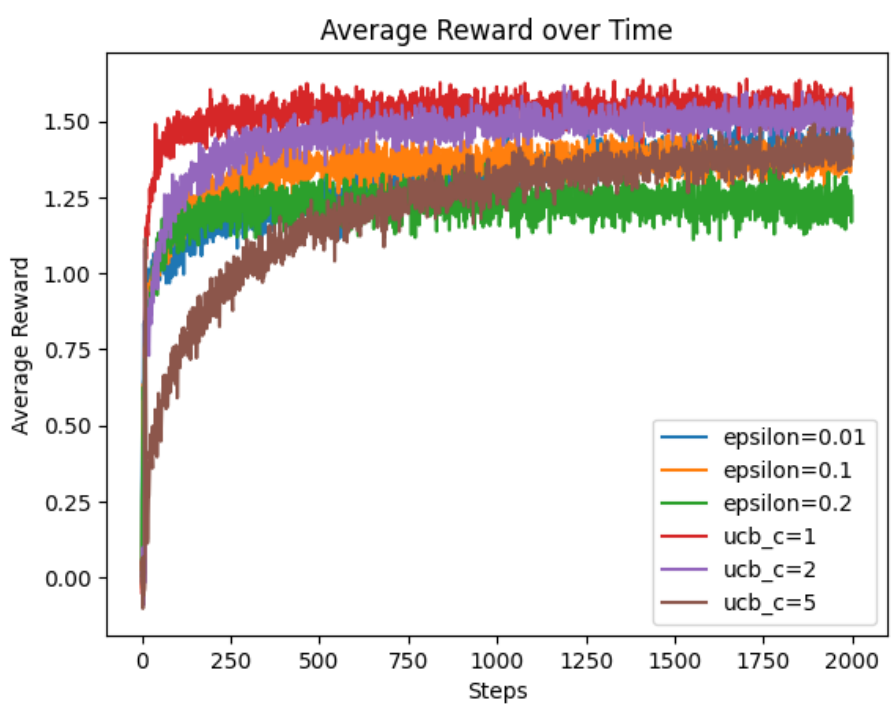

Figure 1: UCB achieves higher average reward over time compared to ε-greedy due to more efficient exploration. While ε-greedy continues random exploration, UCB focuses on uncertain actions, leading to faster convergence.

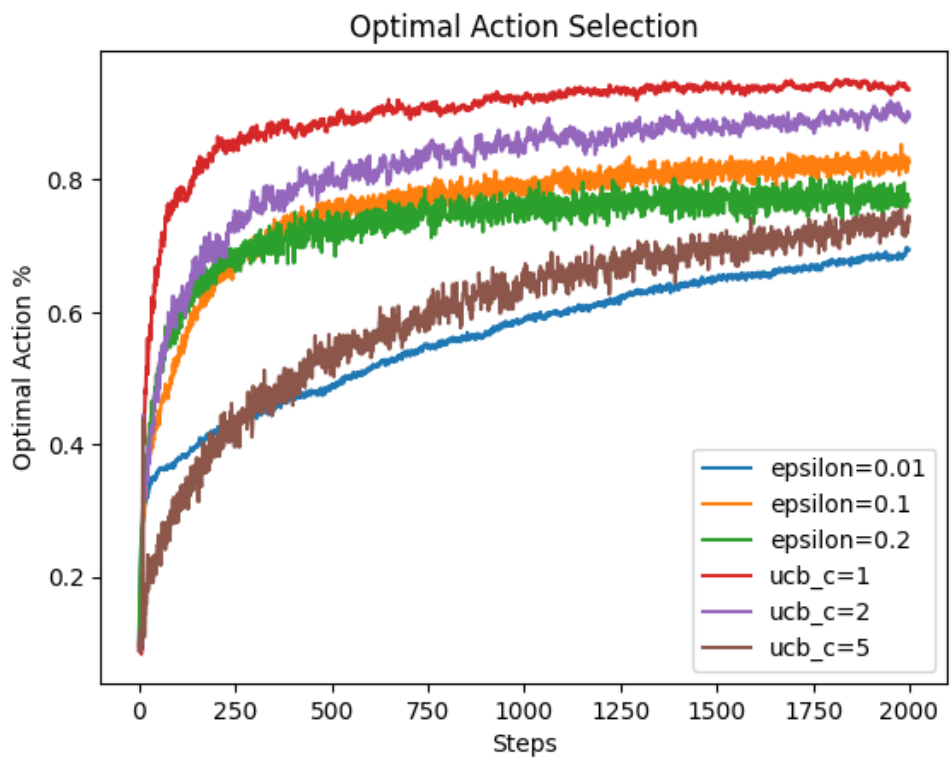

Figure 2: UCB reaches a higher percentage of optimal action selection more quickly, demonstrating its ability to identify the best action faster than ε-greedy methods.

Analysis

The results demonstrate clear differences in how each algorithm balances exploration and exploitation. UCB consistently achieves higher average rewards and converges more quickly to the optimal action compared to ε-greedy strategies. This aligns with Sutton and Barto's discussion (Section 2.7), where UCB is shown to use "optimism in the face of uncertainty" to guide exploration more efficiently.

In contrast, ε-greedy methods rely on random exploration, which can be inefficient. For example, even after identifying a high-value action, ε-greedy continues to explore randomly with probability ε, which can reduce overall performance. Lower ε values (e.g., 0.01) lead to faster exploitation but risk insufficient exploration, while higher ε values (e.g., 0.2) improve exploration but slow convergence.

The optimal action percentage plots further reinforce this behavior. UCB reaches a higher percentage of optimal actions more quickly, indicating that it identifies and commits to the best action faster. This reflects its ability to prioritize actions with higher uncertainty rather than exploring uniformly at random.

These findings highlight the importance of directed exploration in reinforcement learning. While ε-greedy is simple and effective, UCB provides a more principled approach by incorporating uncertainty into decision-making, resulting in more efficient learning and better long-term performance.

AI Use Reflection

Initial Interaction

I used AI to help me build the bandit environment and implement the ε-greedy and UCB agents. I asked for a simple way to simulate a 10-armed bandit using NumPy. The AI gave me a basic version of the environment and the agents, which I used as a starting point.

Iteration Cycle

Iteration 1: My UCB agent was not exploring all actions. I asked why this was happening, and AI explained that actions with zero counts were causing problems. I fixed this by making sure each action was selected at least once.

Iteration 2: My results looked very noisy and inconsistent. I asked AI how to improve this, and it suggested running more simulations. I increased the number of runs to 1000, which made the results more stable.

Iteration 3: I was unsure if I was calculating the optimal action correctly. I asked AI how to track the best action, and I updated my code to compare the agent's action with the true best arm from the environment.

Critical Evaluation

Some suggestions from AI did not fully work at first, so I had to test and adjust the code. I checked if the results made sense by comparing them with what I expected from the theory.

Learning Reflection

I learned how different exploration strategies affect performance. I also learned that AI is helpful, but I still need to test and understand the code. Next time, I will test each part step by step earlier.

Speaker Notes

• Problem: Multi-armed bandit and exploration vs exploitation tradeoff

• Method: Implemented ε-greedy and UCB agents in a custom environment

• Design choice: Tested different ε values and UCB parameters

• Key result: UCB performed better and learned faster

• Insight: Exploring based on uncertainty is more efficient than random exploration

• Challenge: Stabilizing results required multiple runs

• Connection: This concept can be applied to more complex MDP problems